

The Characteristic Equation

An invariant beneath the representations of a linear recurrence

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Abstract

A linear recurrence with constant coefficients can be presented as a row operator on a triangular array, as a companion matrix on a state vector, as a continuant determinant, as a generating function, and as a continued fraction. We treat these as coordinate systems on a single object rather than as competing accounts. The object is the characteristic equation. We make the convergence explicit through worked examples, exhibit the bidiagonal and tridiagonal matrices that generate the standard combinatorial triangles, and identify the transformation group under which the characteristic equation is the invariant. The matrix presentations are fixed only up to a diagonal gauge that preserves a certain product while leaving its factors free; the symmetry class of a representative, real symmetric, Hermitian, or complex symmetric, is therefore a choice within the gauge rather than a property of the sequence. Parameter sweeps within this framework recover the metallic sequences at successive integer values of a diagonal multiplier, and a closing note identifies the named members of the combinatorial family as a vanishingly small subset of a much larger space of formally consistent recurrences, drawing a structural parallel with the moduli-space framing familiar from physics.

A generating triangle as a matrix action

Fix an infinite lower-triangular array $T(n, k)$, with $n \geq 0$ the row index and $0 \leq k \leq n$ the column index. Read each row as a vector $v_n = (T(n, 0), T(n, 1), \dots)^\top$. A two-term recurrence of the form

$$T(n, k) = \alpha(n, k) T(n-1, k) + \beta(n, k) T(n-1, k-1)$$

is exactly the statement that $v_n = L_n v_{n-1}$, where L_n is the lower bidiagonal matrix

$$L_n = \begin{pmatrix} \alpha(n, 0) & & & \\ \beta(n, 1) & \alpha(n, 1) & & \\ & \beta(n, 2) & \alpha(n, 2) & \\ & & \ddots & \ddots \end{pmatrix},$$

carrying α on the main diagonal and β on the subdiagonal. The array is the orbit of the seed vector v_0 under the operators $L_1, L_2, \dots$, and the entire content of the triangle, T is contained in the two functions α, β together with the seed.

Such a presentation treats the standard triangles of Pascal et al as points in a small parameter space. The binomial triangle takes $\alpha = \beta = 1$ on the seed $e_0 = (1, 0, 0, \dots)$. The Eulerian triangle, whose entries encode Worpitzky's identity $x^n = \sum_k \langle n, k \rangle \binom{x+k}{n}$ [12, 3, 5, 2, 9, 7], takes $\alpha = k + 1$ and $\beta = n - k$. The Stirling triangle of the second kind takes $\alpha = k$ and $\beta = 1$. The Delannoy array, genuinely two-dimensional [1], requires a further ingredient of a bidiagonal operator post-composed with the row accumulator U , the cumulative-sum operator with $U_{kj} = 1$ for $j \leq k$, acting on the all-ones seed. The bidiagonal and Riordan-group organisation of such triangles is developed in [10, 11]. Table 1 records this.

Triangle	$\alpha(n, k)$	$\beta(n, k)$	Accumulator	Seed	Row-sum spine
Binomial (Pascal)	1	1	—	e_0	2^n
Eulerian / Worpitzky	$k + 1$	$n - k$	—	e_0	$n!$
Stirling (2nd kind)	k	1	—	e_0	Bell numbers
Delannoy	1	1	U	$\mathbf{1}$	central Delannoy

Table 1: Four combinatorial triangles as points in the space of bidiagonal data $(\alpha, \beta, \text{accumulator}, \text{seed})$. The operator L_n has α on its diagonal and β on its subdiagonal; U is the row accumulator; $e_0 = (1, 0, 0, \dots)$ and $\mathbf{1} = (1, 1, 1, \dots)$. The triangles are not isolated objects but neighbouring choices of weight.

The binomial and Delannoy triangles share the operator L_n and differ only in the accumulator and seed; the Delannoy array is the binomial triangle composed with cumulative summation on a different starting vector. The Stirling triangle differs from the binomial in a single weight, $\alpha = k$ in place of $\alpha = 1$. The Eulerian triangle alone has both weights varying with position. None of this is visible while the triangles are presented as completed diagrams; all of it is immediate once they are presented as matrix data.

Appendix A lays out each of these triangles explicitly, with its first rows displayed, its generating operator written in the form of the opening recurrence, and a single entry of each computed by hand from that recurrence. A reader meeting some of these objects for the first time may wish to consult the appendix before proceeding.

Stirling as a weighted binomial recurrence

The Stirling case deserves emphasis, because the relationship it bears to the binomial triangle is usually left implicit. The binomial recurrence is

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1},$$

and the Stirling recurrence is

$$S(n, k) = k S(n-1, k) + S(n-1, k-1).$$

These are the same recurrence but for the weight on the first term: Stirling places the column index k where the binomial places 1. The binomial triangle is the constant-weight specialisation of Stirling at $k = 1$, or equally, Stirling is the binomial recurrence with a multiplier the binomial had silently set to unity.

The weighting propagates to the row sums. The binomial row sums are 2^n , the powers of the constant total weight while the Stirling row sums are the Bell numbers [8], that satisfy

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k.$$

Here the binomial coefficients are also the kernel of the recurrence. Each Bell number is the convolution of the entire preceding sequence against a row of the binomial triangle.

A single worked instance makes the convolution visible. Taking $n = 3$ and the initial values

$B_0 = 1, B_1 = 1, B_2 = 2, B_3 = 5$, the recurrence gives

$$B_4 = \binom{3}{0}B_0 + \binom{3}{1}B_1 + \binom{3}{2}B_2 + \binom{3}{3}B_3 = 1 + 3 + 6 + 5 = 15,$$

and the four terms are not algebraic accidents. They are the four block-count strata of the 15 partitions of a four-element set. These being weighted in such a way that the construction of B_4 from earlier Bell numbers is precisely the construction of a partition of $\{1, 2, 3, 4\}$. this being done by selecting which k of the first three elements share a block with element 4 and partitioning the remaining $3 - k$ elements among themselves.

The fifteen partitions stratify visibly when one chooses a concrete instance: the integer $210 = 2 \times 3 \times 5 \times 7$ has four distinct prime factors, and so admits exactly $B_4 = 15$ distinct unordered factorisations into products of subsets of its prime factors. They divide by block count as one would expect from $B_4 = 1 + 7 + 6 + 1$: one factorisation as a single line $\{2 \cdot 3 \cdot 5 \cdot 7 = 210\}$, seven as rectangles such as $\{2, 105\}$ or $\{6, 35\}$, six as cuboids such as $\{2, 3, 35\}$ or $\{5, 7, 6\}$, and one as the four-dimensional “hypercube” $\{2, 3, 5, 7\}$ in which every factor sits alone. The geometry behind “partition of a set” becomes vivid in the divisor lattice, and the recurrence on Bell numbers is the recurrence on these geometric strata.

The triangle that generates Stirling and the triangle that weights the recurrence for its row sums are the same triangle. Pascal appears inside the construction of its own near-relation, twice over: once as the unweighted skeleton of the Stirling recurrence, and once as the convolution kernel for the Bell numbers.

The Bell recurrence has full depth; B_{n+1} depends on every earlier term. Its natural matrix presentation is therefore not bidiagonal but upper Hessenberg, with binomial entries filling the band above the diagonal. The bandwidth of the matrix records the depth of the recurrence, and the hierarchy from binomial through Stirling to Bell is a hierarchy of bandwidth.

One sequence, five matrices, one equation

Restrict now our attention to a single spine of T and a constant-coefficient recurrence, where the structure is cleanest. We know that the shallow diagonals of the binomial triangle sum to the Fibonacci numbers, and the projection of the binomial recurrence onto the shallow diagonal is the Fibonacci recurrence itself:

$$\sum_i \binom{n-i}{i} = \sum_i \left[\binom{n-1-i}{i} + \binom{n-2-(i-1)}{i-1} \right] = F_{n-1} + F_{n-2}.$$

Take this projected sequence, $F_n = F_{n-1} + F_{n-2}$, as the object of study, and ask each representation the same question: what does it say about the number whose powers solve the recurrence?

Iteration. Substituting a pure power $F_n = x^n$ and dividing by x^{n-2} ,

$$x^2 = x + 1.$$

Companion matrix. Writing the recurrence as $v_n = Av_{n-1}$ with

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \det(A - \lambda I) = \lambda^2 - \lambda - 1.$$

The equation is now the characteristic polynomial of a matrix, which is where the term *characteristic equation* originates. Its roots are the eigenvalues that govern the growth; from $\det A = -1$ one obtains Cassini's identity $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ with no induction, since $\det(A^n) = (\det A)^n$.

Continuant determinant. The tridiagonal matrix

$$M_n = \begin{pmatrix} 1 & 1 & & & \\ -1 & 1 & 1 & & \\ & -1 & 1 & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & -1 & 1 \end{pmatrix}$$

has, by cofactor expansion along the last row, $\det(M_n) = \det(M_{n-1}) + \det(M_{n-2})$, again the Fibonacci recurrence, again the same characteristic equation. Where the companion matrix is local and moves through "time", the continuant is global standing still; its determinant bulk performing the whole iteration at once.

Generating function. The ordinary generating function

$$G(x) = \sum_{n \geq 0} F_n x^n = \frac{x}{1 - x - x^2}$$

carries $1 - x - x^2$ in its denominator, the reciprocal polynomial of $x^2 - x - 1$; the poles of G are the reciprocals of the roots.

Continued fraction. The golden ratio $[1; 1, 1, 1, \dots]$ is the fixed point of $x \mapsto 1 + 1/x$, so $x = 1 + 1/x$, hence $x^2 = x + 1$.

We have thus, five representations of differing character, analytic and algebraic, finite and infinite, static and dynamic, all forced to the same quadratic. While the representations vary freely the underpinning equation is invariant.

Gauge freedom and its group

The continuant of the previous section was not canonical. Its determinant depends on the off-diagonal entries only through their product. The matrix

$$M_n(b, c) = \begin{pmatrix} 1 & b & & \\ c & 1 & b & \\ & c & 1 & \ddots \\ & & \ddots & \ddots \end{pmatrix}, \quad \det(M_n) = \det(M_{n-1}) - bc \det(M_{n-2}),$$

reproduces Fibonacci for any pair with $bc = -1$. The choices

$$(b, c) = (1, -1), \quad (2, -\tfrac{1}{2}), \quad (i, i)$$

all satisfy $bc = -1$ and yield the identical determinant sequence. The freedom is a genuine gauge symmetry: conjugating by a diagonal matrix $D = \text{diag}(d_1, d_2, \dots)$ sends $b \mapsto (d_i/d_{i+1})b$ and $c \mapsto (d_{i+1}/d_i)c$, preserving the product bc and, with it, the determinant.

The symmetry class of the representative is therefore a choice within the gauge, not a feature of the sequence. Taking $b = c$ real gives a real symmetric matrix. Taking $b = \bar{c}$ gives a Hermitian matrix. Taking $b = c = i$ gives a complex symmetric matrix that is not Hermitian. All three present Fibonacci, because all three respect $bc = -1$.

To see that it is the invariant bc , and not the matrix silhouette, that determines the sequence, hold the shape and change the product. The choice $b = i$, $c = -i$ is Hermitian and has $bc = +1$. Its continuant satisfies

$$\det(M_n) = \det(M_{n-1}) - \det(M_{n-2}),$$

with characteristic equation $x^2 - x + 1 = 0$ and roots on the unit circle. The determinant sequence is the period-six pattern $1, 1, 0, -1, -1, 0, \dots$. The same matrix outline and the same magnitudes of entries, but a different product, give a different equation, and exponential growth gives way to oscillation. What separates Fibonacci from the period-six sequence is the value of the invariant bc , which is precisely the constant term of the characteristic equation.

This is the structure Klein’s Erlangen Program [6] would recognise. A geometry, in Klein’s organisation, is the study of the properties left invariant by a group of transformations. The transformations here are the diagonal conjugations $M \mapsto DMD^{-1}$, which permute the symmetry classes of the continuant, real symmetric, Hermitian, complex symmetric, while fixing the determinant. The orbit of a continuant under this group is a family of matrices of every symmetry type sharing one determinant sequence. The invariant of the orbit is the characteristic equation. A sequence, on this reading, is neither just a matrix nor a recurrence but an equivalence class of presentations, and the characteristic equation is the label the gauge cannot alter.

Three actions, and what plays which role

The word “matrix” has now appeared in three distinct roles, and because all three involve diagonal objects and column vectors at some point, it is worth separating them explicitly.

The first is the *generative* action of Section 1. The bidiagonal operator L_n acts on a row vector v_{n-1} to produce the next row, $v_n = L_n v_{n-1}$. The vector is a row of actual triangle entries; the operator is the recurrence rule; the orbit of the seed v_0 is the triangle. This action steps forward in the row index n .

The second is the *companion* action of Section 2. For a recurrence of depth d , the state is the d -tuple of the most recent terms and the operator is the $d \times d$ companion matrix; for Fibonacci, $d = 2$ and the operator is the 2×2 matrix A . This action also steps forward, but the vector is now a small window of recent values rather than a row of the triangle, and the operator size is the depth of the recurrence.

The third is the *gauge* action of the present section. The diagonal matrix D does not act on any state vector. It acts on the continuant M itself, by conjugation $M \mapsto DMD^{-1}$, rewriting one operator as a similar operator with the same determinant. Nothing is being stepped forward; one is relabelling the basis of a static matrix.

Two cautions follow. First, the gauge matrix D and the seed vector v_0 are unrelated objects. The seed is the input to a generative iteration; the gauge is a change of basis applied to a fixed operator. They share nothing but the incidental presence of diagonal structure.

Second, the depth of a recurrence surfaces differently in the two stepping pictures, and this is the right way to understand the otherwise puzzling difference in matrix shape between the companion matrix and the continuant. In the companion picture, depth is the *fixed dimension* of a square operator that is reused at every step: a depth- d recurrence has a $d \times d$ companion. In the continuant picture, depth is the *bandwidth* of a matrix whose size grows with the term

index n : a depth- d recurrence gives a banded matrix with d active diagonals, and the matrix is $n \times n$ at the n th term. These is why a tridiagonal continuant (bandwidth governed by depth two) and a 2×2 companion (dimension two) describe the same Fibonacci recurrence.

Metallic matrices: a parameter sweep through named sequences

The gauge analysis identifies the constant term of the characteristic equation, the invariant bc , as the carrier of the sequence's identity within a fixed matrix shape. The linear term, set by the diagonal entries, is a second knob, and turning it generates a structurally illuminating one-parameter family. Take the continuant with diagonal m and off-diagonals i ,

$$M_n(m) = \begin{pmatrix} m & i & & & \\ i & m & i & & \\ & i & m & \ddots & \\ & & \ddots & \ddots & i \\ & & & i & m \end{pmatrix},$$

so that the off-diagonal product is $bc = -1$, as for Fibonacci, but the diagonal has been promoted from one to the integer m . Cofactor expansion gives

$$\det M_n(m) = m \det M_{n-1}(m) + \det M_{n-2}(m),$$

with characteristic equation $x^2 = mx + 1$. At $m = 1$ this is the Fibonacci equation $x^2 = x + 1$, with roots the golden ratio and its conjugate. At $m = 2$ it is $x^2 = 2x + 1$, with roots $1 \pm \sqrt{2}$, the silver ratio and its conjugate; the determinant sequence is 2, 5, 12, 29, 70, 169, ..., the Pell numbers. At $m = 3$ the equation is $x^2 = 3x + 1$ and the sequence is 3, 10, 33, 109, 360, 1189, ..., the bronze family. The pattern continues for every positive integer m , producing what are classically called the *metallic* sequences.

This is worth marking because it is a parameter sweep with an unusual property: every integer value of m lands on a named sequence. The continuant remains complex symmetric for every m , the gauge group acts the same way on all of these matrices, and yet the parameter m moves the recurrence from one classically studied object to the next. The roots of the characteristic equation trace a smooth curve in the complex plane as m varies, with the named sequences appearing at the integer points of that curve. The metallic family is a line through the parameter space of recurrences along which combinatorial fame is, against the general expectation we shall meet below, dense.

Why “hard to vary” is the right test

The multiplicity of representations might seem to threaten the idea that a sequence has a single best explanation. If iteration, companion matrix, and continuant¹ all account equally for Fibonacci, and each can be exchanged for another at will, then by Deutsch's criterion [4], that a good explanation is one that is hard to vary, the easy interchangeability looks like the signature of explanations whose details do no work.

The resolution is that the five are not five explanations but five coordinate systems on one explanation, and the one explanation is hard to vary exactly as the criterion requires. The coefficients of the characteristic equation are not independently adjustable. Move the linear coefficient and Fibonacci becomes Pell, $x^2 = 2x + 1$; move the constant and it becomes Jacobsthal,

¹As well as generating function and continued fraction which we are discussed [here](#).

$x^2 = x + 2$; raise the degree and one reaches Tribonacci, $x^3 = x^2 + x + 1$. In each case every representation moves together, because the eigenvalues, the closed-form bases, the generating-function poles, and the continued-fraction limit are all images of the same roots. No coefficient can be changed in one representation without changing it in all. That rigid coupling is what the phrase “hard to vary” names, and it is a property of the equation, not of any representation of it. Deutsch’s own observation that good explanations, being hard to vary, tend to be unique, applies here in a setting he was not addressing: there is one explanation of Fibonacci, the characteristic equation, and there is an abundance of routes to it.

Interactive companions

A browser tools accompany this article and make the matrix origins of the triangles manipulable directly. The first [tool](#) presents the bidiagonal operator L_n as an editable object: the user sets the weights $\alpha(n, k)$ and $\beta(n, k)$, toggles the row accumulator, chooses the seed, and watches the generated triangle and its spines respond, with the preset data of Table 1 available for the binomial, Eulerian, Stirling, and Delannoy triangles. The tool generates the spines as continuant determinants, displaying the tridiagonal matrix whose determinant produces each named sequence and the symmetry class of the chosen representative. The matrix is the primary object and the array or sequence is the derived one, which is the inversion this article argues for.

Coda

We have then that a combinatorial triangle is the orbit of a bidiagonal operator; the operator’s data place it among its neighbours in a small parameter space. A spine of the triangle, usually met as a list, is generated by several operators of differing kinds, each of which exposes the same characteristic equation. The matrix representations of that equation form an orbit under a diagonal gauge group, and the equation is the invariant of the orbit.

What is recognised first as an object, and then as a process, is finally recognised as an invariant preserved across the processes. The characteristic equation earns its name at the last stage:

The Characteristic equation is what is characteristic of the entire family of representations, the structure that survives translation among them.

A note on what is and is not named

A reader who has followed the parameter-space framing to this point will notice a question the paper has not quite answered. The Stirling triangle was identified as the position-dependent promotion of the binomial recurrence: the weight $\alpha = 1$ becomes $\alpha = k$, and the resulting array counts set partitions, with the Bell numbers as row sums. The natural next question is whether the same move applied to the Delannoy recurrence produces a comparably meaningful object. There are three natural candidates, obtained by promoting each of the three Delannoy weights in turn from a constant to a position-dependent function. All three are formally consistent. All three produce well-defined integer arrays with clean recursive structure; the across-step variant, for instance, has factorial boundary row $k!$ and a row-1 closed form $k!(1 + k + H_k)$ involving the harmonic numbers. None of the three, as far as the author can determine, counts a combinatorial entity that mathematics already cares about. Their combinatorial interpretations exist but are reverse-engineered from the recurrences rather than discovered as answers to counting questions already being asked.

This is worth stating clearly, because it sharpens the claim the paper is making. The named triangles in Table 1 are not the only formally consistent points in the parameter space of bidiagonal recurrences; they are a small subset distinguished by their external connections, by counting objects independently studied in combinatorics or arising in independently motivated distributions. Stirling’s status as a named object is a fact about its match with set partitions, not a fact about its position in the space of position-dependent promotions. The space is much larger than the named subset, and most of it is anonymous.

The contrast between two parameter sweeps presented earlier in the paper makes this concrete. The metallic family of § is a sweep through the diagonal parameter m of the Fibonacci-shaped continuant; every integer value of m lands on a classically named sequence (Fibonacci, Pell, bronze, and so on). The three position-dependent promotions of the Delannoy recurrence considered above are sweeps of a different kind, exchanging a constant for a function of the column index, and none of the three lands on anything classically named. The two cases are equally formally consistent. They differ in whether the parameter direction one has chosen happens to align with questions that mathematics was already asking. The metallic direction does so, repeatedly. The Delannoy directions, as far as one can tell, do not. There is no formal way to predict in advance which directions through the parameter space will be lucky in this regard.

The phenomenon resembles, at the level of structural intuition, the situation in the moduli space of string-theory vacua: a vast parameter space of formally consistent constructions contains a vanishingly small subset corresponding to anything observationally meaningful, and the meaningful subset is selected by external connections rather than by the formalism itself. We do not press the analogy beyond this structural point; the selection mechanism is absent in our case, and the parameter space here is finite-dimensional. But the underlying recognition is the same. Formal consistency is cheap. External meaning is rare. The named members of a combinatorial family are named because of their connections beyond the formalism, and a generalisation that is formally consistent need not, and usually does not, inherit those connections.

A The triangles and their generating recurrences

Each triangle below is displayed in its first rows and then written as an instance of the opening recurrence

$$T(n, k) = \alpha(n, k) T(n-1, k) + \beta(n, k) T(n-1, k-1), \quad (*)$$

with the convention that $T(n, k) = 0$ whenever $k < 0$ or $k > n$, and a stated seed for $T(0, 0)$ and, where needed, the rest of row zero. After each, one entry is computed by hand directly from $(*)$ so that the formula may be seen acting.

A.1 Binomial (Pascal), $\alpha = 1$, $\beta = 1$, seed e_0

$$\begin{array}{cccccccc} & & & & 1 & & & \\ & & & & & 1 & & 1 \\ & & & 1 & & 2 & & 1 \\ & & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \\ 1 & & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\ & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \end{array}$$

As an instance of $(*)$: $\alpha(n, k) = 1$ and $\beta(n, k) = 1$, so

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Worked entry. $\binom{4}{2} = \binom{3}{2} + \binom{3}{1} = 3 + 3 = 6$.

A.2 Eulerian / Worpitzky, $\alpha = k + 1$, $\beta = n - k$, seed e_0

The Eulerian number $\langle n, k \rangle$ counts permutations of $\{1, \dots, n\}$ with exactly k ascents. (The triangle below omits the trailing $\langle n, n \rangle = 0$.)

$$\begin{array}{cccccc} & & & & 1 & \\ & & & & & 1 & & 1 \\ & & & 1 & & 4 & & 1 \\ & & 1 & & 11 & & 11 & & 1 \\ 1 & & 1 & & 26 & & 66 & & 26 & & 1 \end{array}$$

$\langle 6, k \rangle : \quad 1, 57, 302, 302, 57, 1.$

As an instance of $(*)$: $\alpha(n, k) = k + 1$ and $\beta(n, k) = n - k$, so

$$\langle n, k \rangle = (k + 1) \langle n-1, k \rangle + (n - k) \langle n-1, k-1 \rangle.$$

This recurrence is the engine of Worpitzky's identity $x^n = \sum_k \langle n, k \rangle \binom{x+k}{n}$, which expresses the monomial x^n in the basis of shifted binomial coefficients; the Eulerian triangle is thus the change-of-basis matrix from that basis to the monomials. The row sums are $n!$, since the ascents partition all $n!$ permutations. *Entry.* $\langle 4, 1 \rangle = (1 + 1) \langle 3, 1 \rangle + (4 - 1) \langle 3, 0 \rangle = 2 \cdot 4 + 3 \cdot 1 = 11$.

A.3 Stirling, second kind, $\alpha = k$, $\beta = 1$, seed e_0

The Stirling number $S(n, k)$ counts partitions of an n -set into exactly k non-empty blocks.

					1				
					0		1		
				0		1		1	
		0		0		1		3	
		0	1		1		7		6
	0		1		15		25		10
0		1		31		90		65	
									15
									1

As an instance of (*): $\alpha(n, k) = k$ and $\beta(n, k) = 1$, so

$$S(n, k) = k S(n-1, k) + S(n-1, k-1).$$

This is the binomial recurrence of A.1 with the constant weight 1 on the first term replaced by the column index k . The new element added to a partition either joins one of the k existing blocks, giving the $k S(n-1, k)$ term, or forms a new block, giving the $S(n-1, k-1)$ term.

Entry. $S(4, 2) = 2 S(3, 2) + S(3, 1) = 2 \cdot 3 + 1 = 7$.

A.4 Bell numbers as a binomial convolution

The Bell number B_n is the total number of partitions of an n -set, hence the row sum $B_n = \sum_k S(n, k)$ of A.3:

$$B_0, B_1, B_2, \dots = 1, 1, 2, 5, 15, 52, 203, 877, \dots$$

The Bell numbers do not satisfy a fixed-depth two-term recurrence and so are not the spine of a bidiagonal operator. They satisfy instead the full-depth binomial convolution

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k,$$

in which the kernel is a row of the binomial triangle A.1. This is the strong sense in which the Bell numbers are recursive-binomial: each term is the binomial-weighted sum of all previous terms. The natural matrix presentation is upper Hessenberg, with binomial entries above the diagonal, the bandwidth recording the full depth of the recurrence.

Worked entry. $B_4 = \binom{3}{0} B_0 + \binom{3}{1} B_1 + \binom{3}{2} B_2 + \binom{3}{3} B_3 = 1 + 3 + 6 + 5 = 15$.

Metallic Bell sequences. The binomial-convolution recurrence above admits a one-parameter generalisation in close parallel to the metallic continuants of §. Multiplying the convolution kernel by m^k at column k ,

$$a_m(n+1) = \sum_{k=0}^n \binom{n}{k} m^k a_m(k), \quad a_m(0) = 1,$$

gives the standard Bell numbers at $m = 1$, the silver Bell sequence

$$1, 1, 3, 17, 179, 3489, 127459, 8873137, 1195313043, \dots$$

at $m = 2$, and the bronze Bell sequence

$$1, 1, 4, 43, 1279, 108472, 26888677, \dots$$

at $m = 3$. Each integer m produces a sequence appearing in the Online Encyclopedia of Integer Sequences, so the parameter sweep lands on documented objects at every integer value, in the same way the metallic continuants do for sequences satisfying $x^2 = mx + 1$.

Worked entry. For $m = 2$,

$$a_2(3) = \binom{2}{0} \cdot 1 \cdot 2^0 + \binom{2}{1} \cdot 1 \cdot 2^1 + \binom{2}{2} \cdot 3 \cdot 2^2 = 1 + 4 + 12 = 17.$$

Each metallic Bell sequence also admits a triangle presentation analogous to the original Bell triangle. The row construction rule is: the first entry of a row is the last entry of the row above, and subsequent entries are formed as m times the entry to the left plus the entry above the left neighbour. The silver case ($m = 2$) reads

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & & 1 & 3 & \\ & & & 3 & 7 & 17 & \\ & & 17 & 37 & 81 & 179 & \\ & 179 & 375 & 787 & 1655 & 3489 & \\ 3489 & 7157 & 14689 & 30165 & 61985 & 127459 & \end{array}$$

and the bronze case ($m = 3$) reads

$$\begin{array}{cccccccc} & & & & & & 1 & \\ & & & & & & 1 & 4 \\ & & & & 4 & 13 & 43 & \\ & & 43 & 133 & 412 & 1279 & & \\ & 1279 & 3880 & 11773 & 35731 & 108472 & & \\ 108472 & 326695 & 983965 & 2963668 & 8926735 & 26888677 & & \end{array}$$

The last entries of successive rows are the sequence values, recovering the metallic Bell sequence in each case. The Bell triangle of the standard $m = 1$ case is the same construction with m set to one, and arises as the row-sum spine of A.3 in the manner described above.

A.5 Delannoy, $\alpha = 1$, $\beta = 1$, accumulator U , seed 1

The Delannoy number $D(n, k)$ counts lattice paths from $(0, 0)$ to (n, k) using steps east, north, and north-east. It is genuinely two-dimensional and is most naturally displayed as a square array rather than a triangle.

	0	1	2	3	4	5	6
0	1	1	1	1	1	1	1
1	1	3	5	7	9	11	13
2	1	5	13	25	41	61	85
3	1	7	25	63	129	231	377
4	1	9	41	129	321	681	1289
5	1	11	61	231	681	1683	3653
6	1	13	85	377	1289	3653	8989

The native recurrence is the three-term relation

$$D(n, k) = D(n - 1, k) + D(n, k - 1) + D(n - 1, k - 1),$$

whose extra term $D(n, k - 1)$ refers to the same row being filled. Writing L for the binomial bidiagonal operator of A.1 and U for the row accumulator $(Uw)_k = \sum_{j \leq k} w_j$, this same recurrence factors as $v_n = ULv_{n-1}$ on the all-ones seed $v_0 = \mathbf{1}$, which is the form recorded in Table 1: the Delannoy array is the binomial triangle post-composed with cumulative summation. The main diagonal $D(n, n)$ gives the central Delannoy numbers 1, 3, 13, 63, 321, 1683, 8989, ... *Entry*. $D(2, 2) = D(1, 2) + D(2, 1) + D(1, 1) = 5 + 5 + 3 = 13$.

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